

Multiperson Respiration Detection: A Digital Programmable Metasurface Analysis Approach

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Abstract—The rapid development of wireless sensing technology has opened new possibilities for noncontact health monitoring. Among them, respiration is crucial information for assessing vital signs. However, traditional methods face challenges of signal interference and overlap in multiperson environments. In this work, we propose a novel method for multiperson respiration detection using a digital programmable metasurface (DPM). This method takes advantage of the modulation characteristics of the DPM in the time and space domain. It divides the channel state information (CSI) from the Wi-Fi transmitter into multiple subsignals in the time domain and modulates the radiation directions of the subsignals for space redistribution. These signals are received by the Wi-Fi receiver and recombined to restore the CSI in each direction, thus enabling the accurate extraction of respiration signals from targets in different directions. Experimental results show that this method can directionally sense the human respiration information in a specific direction under the static working mode. Under dynamic scanning mode, it can effectively separate and detect the respiration information of four people from different directions. This system has great

potential for applications in wireless communication, healthcare, and smart home environments.

Index Terms—Digital programmable metasurface (DPM), respiration detection, Wi-Fi sensing.

I. INTRODUCTION

WITH the development of 5G, 6G, and the Internet of Things (IoT), wireless sensing technology has rapidly become an important foundation for the future intelligent society [1], [2], [3], [4]. This technology utilizes existing wireless networks and adopts advanced methods, such as radio-frequency identification (RFID) [5], [6], [7], ultrawideband (UWB) [8], [9], [10], Wi-Fi sensing [11], [12], [13], and millimeter-wave radar [14], [15], [16]. Without directly contacting the targets, it extracts information by capturing and analyzing the reflection and scattering of wireless signals [17], [18], [19], [20]. By leveraging the propagation characteristics of wireless signals, wireless sensing can detect and monitor the status and activities of objects in the environment without acquiring images or sounds, providing great flexibility and enhancing privacy protection. Wireless sensing technology transcends the limitations of traditional sensing methods, heralding a new era of the intelligent society characterized by cost reduction, increased flexibility, strong privacy protection, and precise sensing capabilities.

Respiration is a fundamental physiological process, and accurately monitoring respiratory patterns is crucial for diagnosing and managing various health conditions. Traditional respiratory detection methods, such as wearable devices and contact-based sensors, often face limitations in terms of comfort and accessibility [21], [22], [23]. Wireless sensing technology offers an alternative approach [24], [25], [26], [27]. By detecting the subtle fluctuations in signals caused by chest expansion and contraction during respiration through wireless signals, the respiratory frequency can be estimated from the scattered signals of the body, enabling remote monitoring without physical contact with the observer. UWB, radar, and RFID have been used to achieve noncontact respiratory sensing [28], [29], [30], [31]. However, these methods usually require specialized sensing devices and are relatively expensive. With the widespread popularity of Wi-Fi, Wi-Fi-based respiratory detection technology has attracted extensive attention [32], [33], [34], [35], [36], [37]. The detection task can be accomplished by analyzing the received signal strength indicator (RSSI) or channel state information (CSI) in existing

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Wi-Fi networks. Among them, Fullbreath applies the conjugate multiplication of CSI between two Wi-Fi antennas to eliminate phase offsets and restore accurate respiratory signals [32]. Farsense proposes a method that uses the ratio of two antennas to break through the range limit of Wi-Fi-based respiratory sensing, significantly increasing the range of Wi-Fi respiratory perception [33]. Wi-Cyclops can capture the CSI changes caused by respiratory movements through a single antenna on a commercial Wi-Fi device for human respiratory sensing in indoor environments [34]. However, the above-mentioned methods can only be implemented in the presence of a single target. In a multitarget environment, the respiration signals from different individuals interfere with each other, making it difficult to accurately distinguish and extract the respiration of each target. Moreover, in real life, as a common wireless device, the Wi-Fi terminal is usually placed in complex living areas, and the number of people in its environment is often random.

Considering the multiperson respiratory detection scenario in complex channel conditions, TinySense uses the time-of-arrival of CSI to filter out the effects of multipath and realizes the respiratory detection of two people [35]. Multisense successfully models multiperson respiratory sensing as a blind source separation problem and simultaneously detects the respiratory information of four people [36]. Based on the IEEE 802.11 standard, Xiong et al. extract respiratory patterns from the channel impulse response (CIR) phase, and simulations verify the centimeter-level positioning accuracy and respiratory perception ability of this method in multiperson scenarios [37]. However, the above-mentioned methods often require more complex algorithms or the usage of large-scale antenna arrays to achieve, which limits the promotion of the system in practical applications.

As a revolutionary 2-D structure, metasurface has emerged as a cutting-edge electromagnetic (EM) wave manipulation technology. By designing subwavelength artificial structures, metasurfaces can precisely control EM wave properties, such as amplitude, phase, and polarization [38], [39], [40], [41], [42], [43]. Additionally, digital programmable metasurface (DPM) have overcome the limitations of fixed-function designs in manufacturing [44], [45], [46]. DPMs utilize digital coding to represent the different EM responses of the meta-atom. By changing the digital coding, flexible and rapid switching between each state can be achieved. This innovation has paved the way for dynamic real-time manipulation of EM waves and direct information processing, giving rise to applications, such as reconfigurable beamforming, information entropy control, wireless communication, and imaging [47], [48], [49], [50], [51], [52].

With the help of metasurface technology, the application scenarios in wireless environments have been significantly expanded, especially in the field of wireless sensing. In the field of respiratory sensing, researchers have accurately captured the respiratory signals of multiple targets by fully exploiting the space-time modulation characteristics of metasurfaces [53], [54]. However, existing methods face several challenges: the optimization algorithms involved are often

complex, and the system stability may be compromised in practice. Moreover, the generation of harmonic signals can interfere with regular wireless channels, necessitating a tradeoff between sensing performance and communication quality in practical deployments. Current time-space modulation mechanisms rely on single-tone excitation signals, using the resulting harmonics to sense multiple targets from different directions. Yet this approach has critical limitations. First, single-frequency excitation inherently produces harmonic frequencies that pollute the spectrum and interfere with existing wireless networks, rendering such methods incompatible with current communication platforms. Second, they require complex real-time optimization to scan the spatial domain across multiple harmonic orders, leading to high computational overhead and stringent hardware demands. Additionally, energy dispersion across multiple harmonics results in low energy efficiency. These limitations underscore the need for more spectrum-friendly, computationally lightweight, and energy-efficient solutions to support scalable and practical wireless sensing applications.

To address the above-mentioned issues, this article presents a multiperson respiration detection method that applies the DPMs' flexible modulation abilities for EM waves in both time and space dimensions. A typical application scenario is shown in Fig. 1. This method takes advantage of the space-time modulation characteristics of DPM to divide the CSI from the Wi-Fi transmitter into multiple subsignals in the time domain and modulate the radiation directions of the subsignals for spatial redistribution. Upon interacting with the human targets in the environment, these signals are received by the Wi-Fi receiver and recombined to restore the CSI in each direction. This enables the accurate extraction of respiration signals from targets in different directions. Practical experiments have verified that, under static working mode, this method can detect the respiration information of targets in a specific direction while filtering out interference from other targets. In the dynamic scanning mode, it can concurrently separate and extract the respiration signals of four individuals. The results show that, without modifying the Wi-Fi structure, this method can effectively separate the respiration features of multiple targets.

The main contributions of this article are summarized as follows.

- 1) We propose a CSI space-time slicing and recombination method based on DPM. By leveraging the space-time modulation characteristics of DPM, the CSI from the Wi-Fi transmitter are divided into multiple subsignals in the time domain and redistributed in space. Upon interacting with the targets, these signals are received by the Wi-Fi receiver and recombined to restore the CSI in each direction, thereby enabling the precise extraction of respiration signals from targets in different directions.
- 2) We design a DPM with 2-bit phase-modulation functionality in the Wi-Fi frequency band. By controlling the reflection coefficients of its column meta-atoms through digital coding, this DPM can manipulate the Wi-Fi signal to perform single-beam scanning within the $\pm 60^\circ$ interval, significantly enhancing the signal strength



Fig. 1. Typical application scenarios of multiperson respiration detection based on DPM-aided Wi-Fi. The DPM works with Wi-Fi terminal to separate and detect breathing information from multiple people in the same space using time and space modulation capabilities.

in the designated direction and reducing the scattering energy in other directions.

- 3) Based on the designed DPM and the standard Wi-Fi transceiver architecture, we construct a multitarget respiration sensing system based on DPM. The system features two operating modes in multiperson scenarios: a) static operating mode and b) dynamic scanning mode. Experiments have verified that the DPM-aided Wi-Fi sensing system can directionally enhance the respiratory signals in the target direction while reducing interference from other directions, thus enabling the effective extraction of multiperson respiratory signals.

The structure of this article is as follows. In Section II, we briefly analyze the limitations of existing Wi-Fi-based respiration sensing technologies, focus on discussing the interference problems in multiperson scenarios under the multipath model, and introduce DPM to enhance the scattering energy of specific targets. In Section III, based on the space-time modulation characteristics of DPM, a CSI space-time slicing and recombination method based on DPM is proposed. A prototype of a DPM-aided Wi-Fi sensing system is then presented, accompanied by the design of a 2-bit DPM tailored to operate within the Wi-Fi frequency band for this system. In Section IV, the comprehensive performance of DPM is validated within an anechoic chamber. Utilizing the designed DPM and the standard Wi-Fi transceiver architecture, the prototype system is constructed and tested in a real-world scenario. The experimental results are meticulously analyzed thereafter. Finally, Section V offers a comprehensive summary of the work presented in this article.

II. ANALYSIS OF LIMITATIONS OF WI-FI RESPIRATION DETECTION

In recent years, Wi-Fi signals have been widely used to monitor health conditions, such as human respiration. Among them, Wi-Fi CSI reflects various changes that the Wi-Fi signal experiences during propagation, such as amplitude attenuation, phase shift, and time delay. Under ideal conditions, the CSI signal $H(f)$ with a central frequency of f can be given as

$$H(f) = Ae^{-j2\pi \frac{d}{\lambda}} \quad (1)$$

where A is the complex attenuation, d is the length of the propagation path, and λ is the wave length. Fig. 2(a) shows the schematic of Wi-Fi-based respiration detection. The rhythmic expansion and contraction of the chest during respiration lead to changes in the amplitude and phase characteristics of CSI. By extracting the CSI and using the variational mode decomposition (VMD) algorithm, the respiration signal can be recovered from the Wi-Fi channel information. The Lagrange function $L(\{u_k\}, \{\omega_k\}, \lambda)$ constructed by the VMD algorithm can be expressed as

$$L(\{u_k\}, \{\omega_k\}, \lambda) = \alpha \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 + \left\| h(t) - \sum_{k=1}^K u_k(t) \right\|_2^2 + \langle \lambda(t), h(t) - \sum_{k=1}^K u_k(t) \rangle \quad (2)$$

where u_k is the k th modal component, which is the signal component obtained by decomposition; ω_k is the center frequency

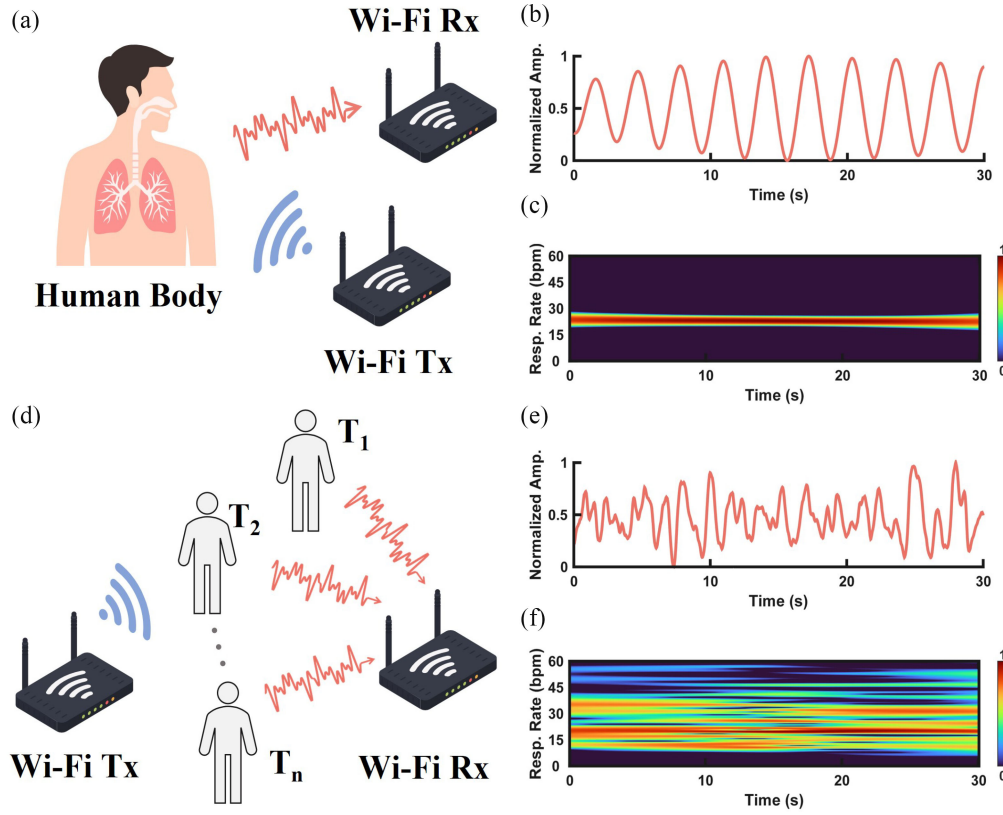


Fig. 2. (a) Schematic of Wi-Fi single-person respiration detection. (b) Time-domain signal of an individual's respiration extracted from Wi-Fi CSI. (c) Spectrogram of an individual's respiration extracted from Wi-Fi CSI. (d) Schematic of Wi-Fi multiperson respiration detection. (e) Time-domain signal of multiperson respiration extracted from Wi-Fi CSI. (f) Spectrogram of multiperson respiration extracted from Wi-Fi CSI.

corresponding to the k th modal component; $h(t)$ is the original time-domain CSI as the signal to be decomposed; $\lambda(t)$ is the Lagrange multiplier; $h(t) - \sum u_k(t) = 0$ represents the penalty for the constraint; α is the smoothing parameter, controlling the smoothness of the decomposed modes; ∂_t represents the derivative to time t , calculating the frequency characteristics of the signal; $\delta(t) + (j/\pi t)$ is the Hilbert transform kernel, used to obtain the analytic form of the signal; $\|\cdot\|_2^2$ is the square of the norm. The objective of this Lagrange function is to decompose the CSI data received by Wi-Fi into k -modal components u_k and ensure that the sum of the decomposed modes is as close as possible to the original signal. After the CSI is processed by the VMD algorithm, the time-domain signal of normal human respiration is shown in Fig. 2(b). It depicts the stability of the respiration rate in the time domain. However, the above-mentioned respiration detection scenario is under ideal conditions. In practical applications, due to the multipath effect, each path will experience different degrees of amplitude attenuation, phase shift, and time delay, resulting in signal overlap at the receiver. In addition, when there are multiple people in the environment, the respiration signals from different people will interfere and overlap, and the finally received CSI can be expressed as

$$H(f) = \sum_{q=1}^Q A_q e^{-j2\pi \frac{d_q}{\lambda}}. \quad (3)$$

In the formula, Q is the total number of propagation paths, and A_q and d_q are the complex attenuation and path length under

the q th path, respectively. When there are multiple people, the signal emitted by the Wi-Fi Tx will be reflected by different people and received by the Wi-Fi Rx, as shown in Fig. 2(d). In this case, the respiration time-domain signal recovered from the CSI is shown in Fig. 2(e). Due to the interference and overlap of respiration signals from different individuals, the time-domain waveform becomes chaotic, making it difficult to distinguish the respiration waveforms of individual targets. Fig. 2(f) shows the respiration spectrum in a multiperson scenario. Generally, the normal respiration frequency of an adult is 12–20 breaths per minute (bpm). However, due to the superposition of respiration signals from multiple individuals, it is impossible to distinguish the respiration information of each individual from the same time–frequency diagram. Second, the aliasing of respiration signals with different frequencies leads to the generation of combined frequencies, contaminating the time–frequency spectrum. In Fig. 2(f), nonnormal respiration frequencies can be observed, and it is impossible to judge and separate the respiration characteristics of different targets.

It can be seen that the Wi-Fi-based, multiperson respiration sensing results in a complex multipath environment that is unreliable. The fundamental reason is that the respiration signals of each individual are reflected by multiple paths, interfere with and superimpose on each other, and then are received by the Wi-Fi receiver. The Wi-Fi signals are uncontrollable. Here, we introduce the DPM to artificially manipulate the reflection paths of EM waves. By dynamically regulating the working states of each meta-atom on the DPM, customized manipulation of EM signals in time and space can be achieved.

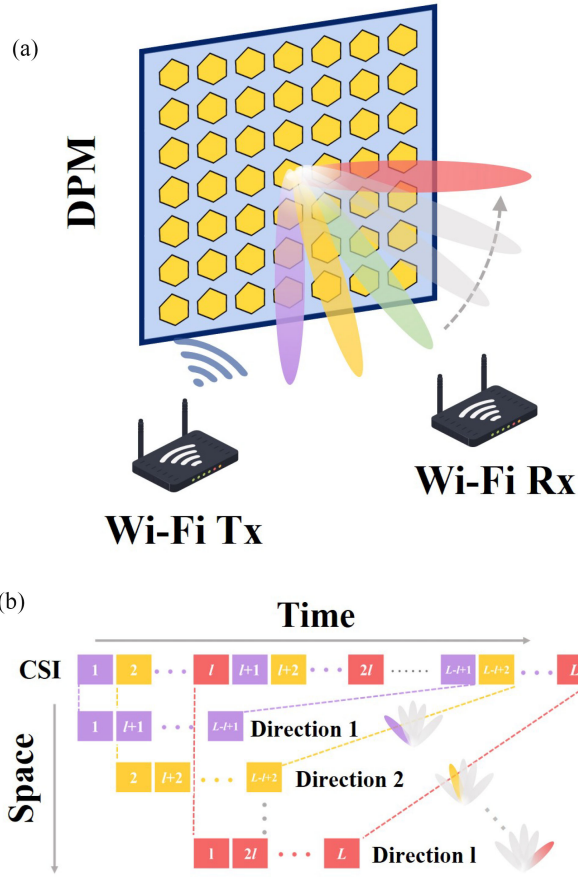


Fig. 3. (a) Schematic of the DPM-aided Wi-Fi system. (b) Schematic of CSI time-space slicing and recombination.

We propose a new respiration sensing method based on DPM to solve the problems of multipath interference and superposition in Wi-Fi sensing and realize the simultaneous sensing of multiperson respiration signals.

III. PROPOSED DPM-BASED MULTIPERSON BREATHING DETECTION METHOD

A. DPM-Aided Wi-Fi Sensing System Prototype Design

To overcome the interference and mixing of respiration signals from multiple individuals, we propose a cooperative framework between DPM and Wi-Fi terminals to slice and recombine CSI across both temporal and spatial dimensions. Specifically, the DPM is a 2-D array composed of reprogrammable meta-atoms embedded with adjustable devices. In the DPM, the reflection phase of the meta-atom can be switched by applying different control voltages. The reflection phase of each meta-atom can be periodically switched according to the corresponding digital binary. For the 1-bit case, the binary representation is “0/1,” while for the 2-bit case, it is “00/01/10/11.” Here, when a plane wave is incident normally, the far-field function scattered by the DPM can be expressed as

$$f(\theta, \varphi) = \sum_{m=1}^M \sum_{n=1}^N E_{mn}(\theta, \varphi) \Gamma_{mn} \times e^{-j \frac{2\pi d_{\text{meta}}}{\lambda} \sin\theta[(m-1)\cos\varphi + (n-1)\sin\varphi]} \quad (4)$$

where θ and φ are the elevation angle and azimuth angle in an arbitrary direction, respectively, Γ_{mn} is the reflection coefficient of the (m, n) th meta-atom, d_{meta} is the spacing between each meta-atom, λ is the wavelength of the central frequency, and E_{mn} is the meta-atom scattering pattern. By optimizing Γ_{mn} of different meta-atoms, the DPM can achieve different scattering effects, including the directional gain G_T , which can be expressed as

$$G_T = \frac{4\pi |f(\theta, \varphi)|^2}{\int_0^{2\pi} \int_0^{\pi/2} |f(\theta, \varphi)|^2 \sin\theta d\theta d\varphi}. \quad (5)$$

By optimizing different Γ_{mn} , the improvement of beam gain in a specific direction can be achieved. Here, we combine the DPM with the Wi-Fi system. We can enhance the energy of a specific-angle path in a multipath environment. In this case, the CSI in (3) can be modified to

$$H(f) = \sum_{q=1}^{Q-1} A_q e^{-j2\pi \frac{d_q}{\lambda}} + G_T A_T e^{-j2\pi \frac{d_T}{\lambda}} \quad (6)$$

where A_T and d_T are the complex attenuation and path length along the directional path. This formula shows that after the CSI is redistributed by the DPM, the signal energy in the designated direction is enhanced. When the signal energy is increased to a sufficient extent, (6) can be further simplified to

$$H(f) \approx G_T A_T e^{-j2\pi \frac{d_T}{\lambda}}. \quad (7)$$

The signal energy after the DPM beamforming is sufficient to overpower the signals from other angles. This enables the enhancement of signal energy in a specific direction and weakens the interference from signals in other directions.

Based on the proposed theory, we put forward a DPM-aided Wi-Fi respiration sensing system. The prototype design of the system is shown in Fig. 3(a). The signal transmitted from the Wi-Fi Tx is spatially redistributed by the DPM. Then it is directed to different directions before being received by the Wi-Fi Rx. The beams in different colors in Fig. 3(a) correspond to different CSI slices, which are redistributed in the designed direction. As indicated by (7), the extraction of the information of the corresponding CSI slice from the Wi-Fi Rx enables the acquisition of the sensing information in the specific direction and diminishes interference from other directions.

Specifically, we assume that the signal transmitted by the Wi-Fi Tx reaches the DPM. In the time domain, the signal has a total length of L and is composed of sub-CSI structures each with a length of l . Here, l represents the space scanning period of the DPM. Each sub-CSI is further evenly divided into subblocks according to space beams and then spatially distributed in various directions by the DPM, which can be expressed as

$$\text{CSI}[n] = \sum_{k=0}^{K-1} \text{CSI}_k[n] \quad n = k \cdot l, k \cdot l + 1, \dots, (k+1) \cdot l - 1 \quad (8)$$

where $K = L/l$ is the number of sub-CSI blocks. The CSI information in different directions can be further expressed as

$$\text{CSI}_i[n] = \text{CSI}[n] \cdot \text{Beam}_i[n] \quad (9)$$

where $\text{CSI}_i[n]$ represents the CSI signal of the i th direction and $\text{Beam}_i[n]$ is the beam masking function for the i th direction, which can be expressed as

$$\text{Beam}_i[n] = \begin{cases} 1, & \text{if } n \in \text{time slot of direction } i \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

The segmentation result is shown in Fig. 3(b). In the space domain, the CSI received at the receiving end undergoes down-sampling. From the received CSI, the CSI signal in each direction can be separately extracted. Compared with the original CSI, the sampling rate becomes $1/l$ of the original. Leveraging the space-time modulation capability of the metasurface, the receiver first performs down-sampling on the fully sampled CSI signals. This down-sampling is conducted according to the time cycles allocated by the metasurface in the space domain. Specifically, it extracts CSI signals for each directional target through time-division multiplexing and can be expressed as

$$\text{CSI}_i^{\text{down}}[m] = \text{CSI}_i[m \cdot l] \quad m = 0, 1, \dots, \frac{L}{l} - 1 \quad (11)$$

where $\text{CSI}_i^{\text{down}}[m]$ represents the downsampled signal for i th direction, and the sampling rate is reduced to $(1/l)$ of the original rate. The proposed method applies the VMD algorithm to these downsampled CSI signals from different directions, enabling effective extraction of respiratory information.

As long as the Nyquist sampling theorem is satisfied, the respiration signals from each direction can be accurately reconstructed. For example, in Fig. 3(b), the purple CSI block corresponds to a specific spatial direction. During reception, CSI data are downsampled starting from this block at a rate of $1/l$ relative to the original. These downsampled signals are then recombined to reconstruct the CSI for Direction 1. By processing this reconstructed signal, the system can extract sensing information for that direction. Similarly, this approach can be applied to other directions. Compared with traditional Wi-Fi sensing methods, the proposed DPM-aided approach can sense targets in multiple directions within a single CSI. Moreover, during the sensing process, it can minimize interference from other directions as much as possible, enabling efficient multitarget detection.

B. Design of 2-Bit DPM

To implement the proposed DPM-aided Wi-Fi sensing system, we designed a DPM with a 2-bit phase quantization operating in the Wi-Fi frequency band. Fig. 4 shows the schematic of the meta-atom of the proposed DPM. The working principle of this DPM is as follows: When irradiated by a far-field source in the upper half-space (where $z > 0$), alterations to the DPM's coding enable the achievement of four distinct coding states. There is a 90° phase difference between adjacent states, while the amplitude remains unchanged.

As shown in Fig. 4(a), the structure of the DPM meta-atom is presented in a 3-D view, and its polarization direction is the x -direction. The meta-atom structure is stacked using printed circuit board (PCB) technology and consists of three layers: 1) the patch layer; 2) the dielectric layer; and 3) the ground layer. The patch layer is composed of an irregular hexagonal

patch in the middle and feeder lines on both sides, which are connected through two PIN diodes (MADP-000907-14020x). The dielectric layer is F4B with a thickness of $H = 2.57$ mm (dielectric constant of 2.65 and loss tangent of 0.02). The ground layer is an unetched metal patch. A metal via in the center of the hexagonal patch penetrates the three-layer structure, connecting the ground planes of the digital and analog signals. The optimal parameter settings of the structure are as follows: $p = 25$ mm, $w_1 = 12.428$ mm, $w_2 = 10.235$ mm, $w_3 = 9.551$ mm, $w_4 = 15.882$ mm, $w_5 = 9.297$ mm, $w_6 = 9.894$ mm.

Based on the above design, we fabricated a DPM, and the processing result is shown in Fig. 4(c). The entire metasurface contains 8×8 meta-atoms, with an overall size of $150 \text{ mm} \times 23 \text{ mm} \times 2.76 \text{ mm}$. Two PIN diodes are soldered to each meta-atom. The PIN diodes of each column of meta-atoms share the same bias voltage, and the bias feeder lines on the same side of each column of meta-atoms are connected, sharing the same digital control signal. At the same time, all the ground lines are connected to the metal back-plane of the ground layer. Finally, all the bias feeder lines and ground lines are integrated into a unified interface, which is connected to the control platform via signal lines.

IV. REAL EXPERIMENT VERIFICATION AND ANALYSIS

To construct the proposed DPM-aided Wi-Fi respiration sensing system, we first tested the EM response of the designed and fabricated 2-bit DPM. By optimizing the digital coding configuration, we evaluated its single-beam scanning performance in space. Subsequently, based on the designed DPM and the standard Wi-Fi system, we built the proposed system. The system we proposed features two working modes: static operating mode and dynamic scanning mode. The static operating mode is used to verify the basic theory and extract the respiratory information of the target in the specified direction at the maximum sampling rate. The dynamic scanning mode reorganizes the CSI time slices and space information, achieving the effect of multitarget detection.

A. Performance Measurement of 2-Bit DPM

To validate the performance of the designed 2-bit DPM, we tested its EM response and beam-scanning performance in a standard anechoic microwave chamber. The detailed experimental setup is illustrated in Fig. 4(d). A linearly polarized horn antenna operating at 4–8 GHz is placed along the normal direction of the metasurface. They are fixed on a turntable at one end of the anechoic chamber together with the DPM, and they can rotate 360° in the horizontal plane to facilitate the measurement of the scattering pattern. At the other end of the anechoic chamber, a horn antenna with the same linear polarization is fixed to receive the scattering pattern in the horizontal plane.

First, we test the reflection coefficient of the DPM using the free-space method. Horn antenna 1 is connected to a vector network analyzer (Agilent N5245A), and its polarization direction is the same as that of the DPM. By adjusting the digital output voltage through the control platform, we controlled the on/off states of the PIN diodes, respectively, to test the four

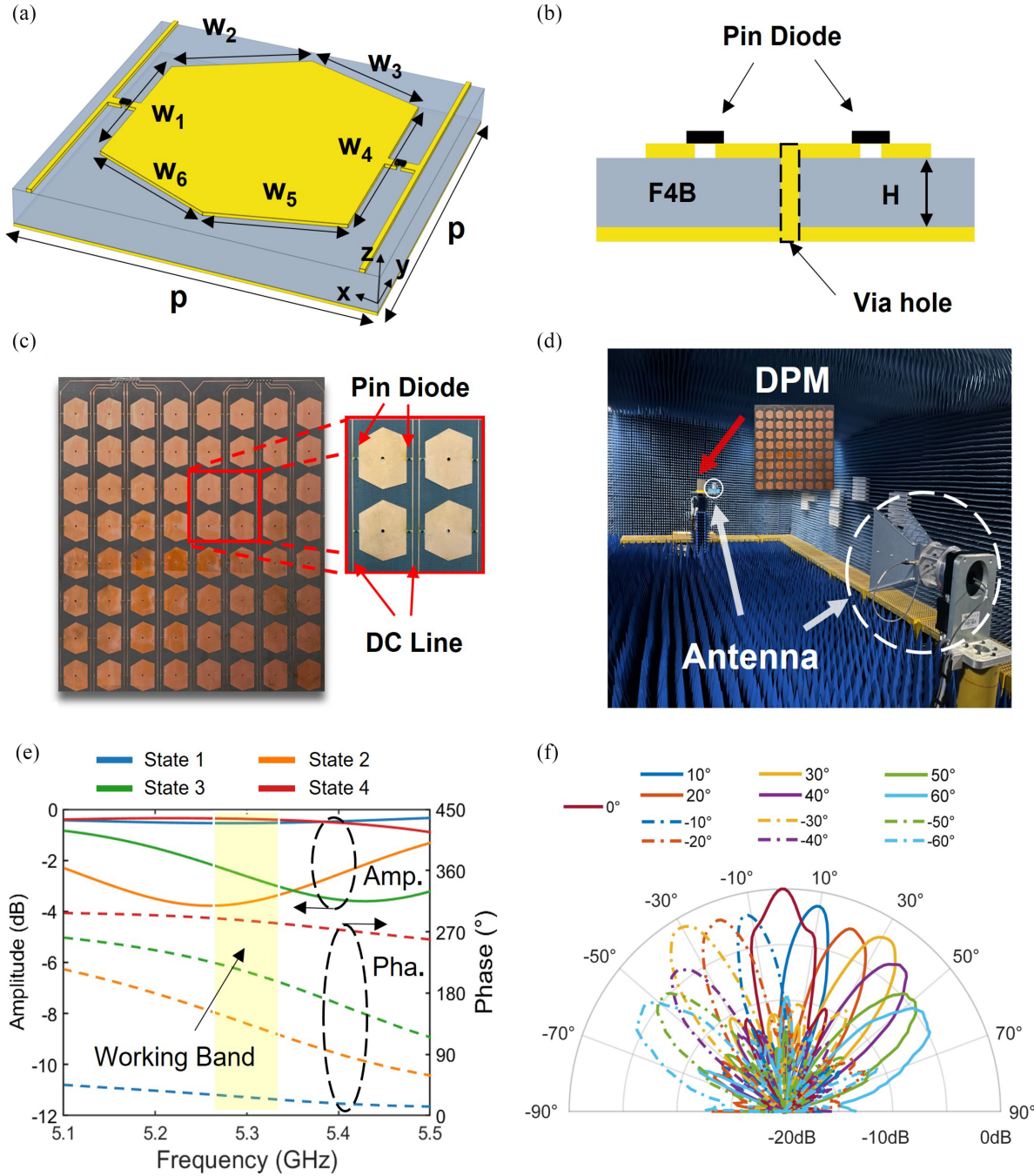


Fig. 4. (a) Meta-atom structure of the DPM. (b) Stacked structure of the DPM meta-atom. (c) Reflection amplitudes and phases of the DPM in four discrete states. (d) Scattering patterns of the DPM with normal incidence in diverse coding states.

states of the DPM. Additionally, depending on the position of the DPM, we set a corresponding time-domain gate in the received signal of the vector network analyzer to filter out the influence of environmental noise on the test results. Fig. 4(d) shows the reflection amplitudes and phases of the DPM in four states within the 5.1–5.5 GHz frequency band. The operating frequency band of this DPM is $5.3 \text{ GHz} \pm 20 \text{ MHz}$, and there is a phase difference of $90^\circ \pm 10^\circ$ between its four states, meeting the design requirements for operating in the Wi-Fi frequency band. The amplitudes of state 2 and state 3 are relatively low, which may be caused by two factors leading to

the nonideal low-reflection amplitudes: 1) energy dissipation of the F4B substrate due to its high-loss tangent of 0.02; and 2) in the actual fabricated structure, there is no isolation design between the digital and analog feeding networks, and part of the energy will be coupled into the feeding network.

Subsequently, based on the measured reflection coefficients of the DPM and in combination with the EM scattering model presented in (4), we optimize the beam coding of the DPM in the range from -60° to 60° (with an angle interval of 10°) using the genetic algorithm. Similarly, we conduct the test in the anechoic microwave chamber. By

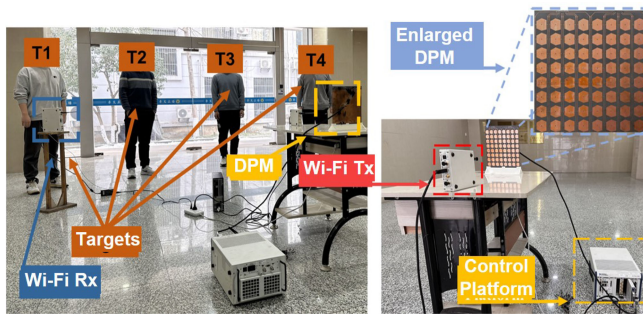


Fig. 5. (a) Experimental scenario for multiperson respiration detection, including four volunteers standing at different angles and the DPM-aided Wi-Fi system. (b) DPM with DIO lines and a control platform.

converting the optimized coding into control voltages, the control platform inputs them to the feeding ports through DIO lines to regulate the coding states of each meta-column. During the test, antenna 1 and antenna 2 are connected to port 1 and port 2 of the vector network analyzer, which is used to provide the feed signal to the DPM and receive the far-field scattering signal, respectively. After performing angle scanning and recording data in the range from -60° to 60° , the results are shown in Fig. 4(f). The results show that the beam gain of the DPM is around 10 dB in -60° to 60° . However, as the scanning angle increases, the beam width gradually increases and the gain decreases accordingly. This phenomenon might lead to a reduction of the sensing performance at large angles, with the SNR degrading in comparison to that at small angles. Therefore, this performance change needs to be fully considered in practical applications.

B. Hardware and Software Architecture of DPM-Aided Wi-Fi System

We constructed a DPM-aided Wi-Fi transceiver system using the designed 2-bit DPM. This system consists of a pair of Wi-Fi transceivers, the DPM, and its control system. The Wi-Fi transceiver system employs the PicoScenes software-based Wi-Fi signal transceiver platform on USRP, which features easy operation, packet injection, and software-based baseband implementation, meeting the actual testing requirements of Wi-Fi scenarios [55]. PicoScenes is a public platform capable of extracting CSI from 802.11ax frames. This feature is highly consistent with the operational logic of commercial Wi-Fi devices in terms of data processing and signal analysis. As a result, it ensures that the signal interaction and data collection methods in the test environment are identical to those in the actual commercial environment. The DPM is a custom-designed and fabricated one with 2-bit phase modulation capability.

The DPM-aided Wi-Fi system's hardware and software architecture comprises three key components. The 2-bit DPM, which supports discrete phase modulation, is fabricated and connected to a control system via DIO lines. The control system is based on an NI PXIe-1082 chassis, with the NI PXIe-8881 central controller running Labview for task scheduling and data storage, communicating with peripheral modules via the PXI Express bus. An NI PXIe-6674T timing module

manages synchronization, generating submicrosecond clock signals for precise alignment between the USRP and DPM control system, while the NI PXIe-6581B DIO module outputs 0/1.4V control signals through DIO lines to modulate metasurface. On the FPGA side, LabVIEW FPGA programming on the NI PXIe-7966R maps the metasurface columns to I/O pins, with a hardware timer generating microsecond-precision phase switching sequences and a lookup table converting digital codes to analog voltages for 2-bit phase modulation.

The signal processing workflow is implemented in MATLAB. CSI data collected by the USRP is down sampled to extract direction-specific components, decomposed using the VMD algorithm to isolate respiration-related signals, and then analyzed through short-time Fourier transform to estimate respiratory rates. To achieve precise timing control, the PXIe-6674T timing module exchanges trigger signals with the USRP via external cables. Before each Wi-Fi frame transmission, the USRP sends a GPIO trigger to the timing module, which then generates a precision pulse-per-second (PPS) signal to update the DPM phase profile via the FPGA, ensuring strict synchronization between the Wi-Fi transmission/reception and DPM modulation.

The overall schematic of the entire system is shown in Fig. 5. The Wi-Fi Tx is placed along the normal direction of the DPM, and the Wi-Fi Rx is located at a distant site to collect Wi-Fi signals in space. The antenna polarization directions of the two Wi-Fi terminals are the same as the DPM to ensure efficient modulation of Wi-Fi signals. The CSI packet-sending time is set to 30 s. During the experiment, the volunteers are positioned at different orientations around the metasurface and remain stationary throughout.

C. Directional Respiration Detection Under Static Operating Mode

The system operates in two modes: 1) static operating mode and 2) dynamic scanning mode. In the static operating mode, the DPM creates a stable phase gradient on its surface to achieve directional beamforming. In a multiperson scenario, it can enhance the respiration signals of the target person in a specific direction while reducing the strength of respiration signals from other directions. In the dynamic scanning mode, the DPM divides the CSI into multiple subsignals in the time domain. It proceeds to modulate the radiation directions of these subsignals for scanning multiple people in space. Eventually, at the receiving end, the respiration information of distinct persons in each direction can be sequentially retrieved and reconstructed.

First, we tested the DPM in the static operating mode. In specific scenarios, we prefer to pay attention to the person in a certain direction, minimizing interference from others. The experimental setup remained the same as previously described. In the environment, there were only two individuals. One person stood at an angle of 30° relative to the center of the DPM, while the other stood at -20° , and both were positioned 3 m away from the DPM's center. Fig. 6(a) shows the experimental schematic when the DPM is not working. In this state, all meta-atoms of the DPM have

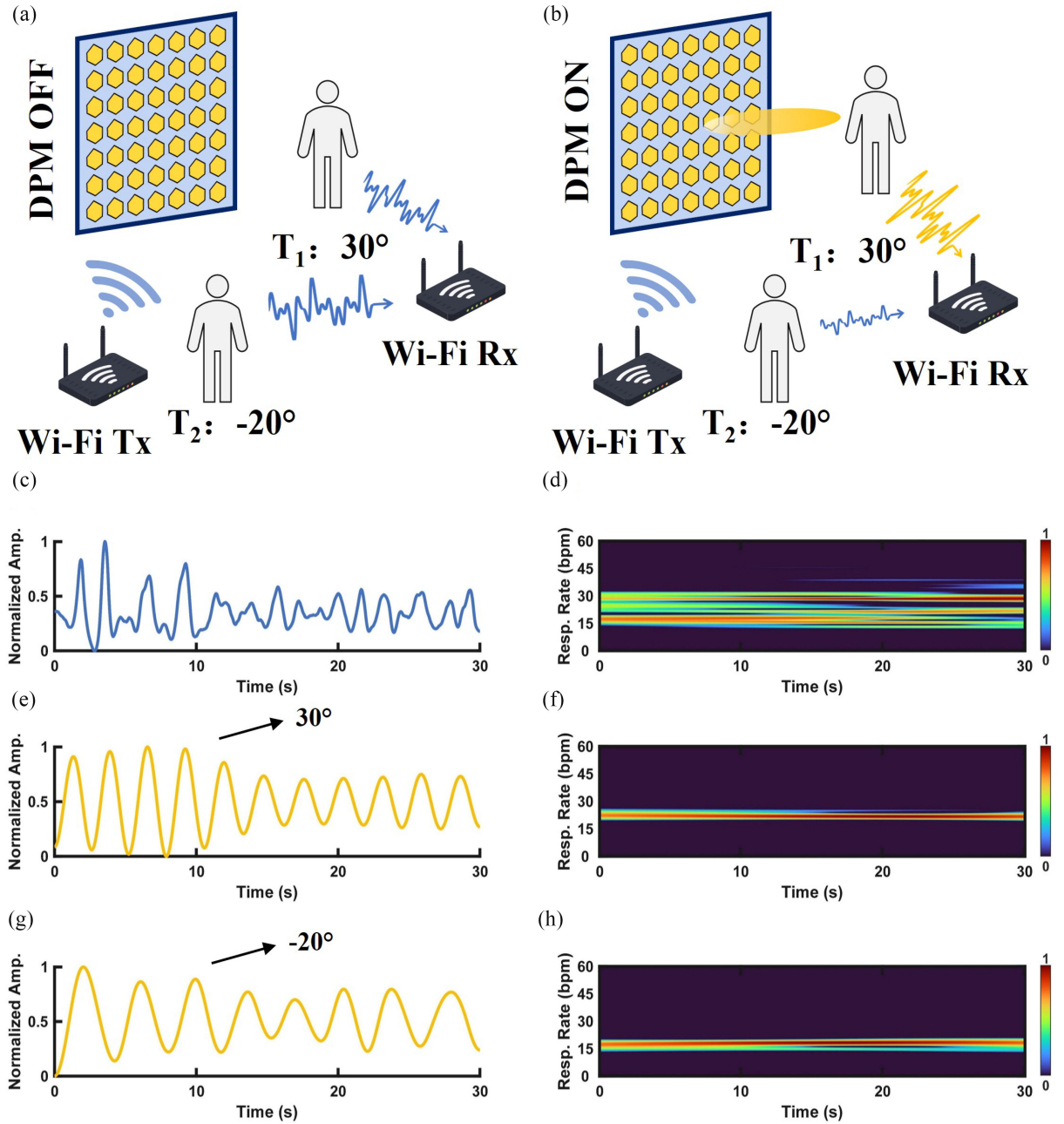


Fig. 6. DPM-aided Wi-Fi respiration detection for two people in a nonoperating state. (a) Is the experimental schematic, and (c) and (d) are the respiration signal and spectrogram extracted from CIS, respectively. DPM-aided Wi-Fi respiration detection for two people in an operating state. (b) Is the experimental schematic when the DPM beam is formed to 30° , (e) and (f) are the respiration signal and spectrogram extracted when the target is at 30° , respectively, and (g) and (h) are the respiration signal and spectrogram extracted when the target is at -20° , respectively.

the same coding state, resulting in no beamforming effect. The measured CSI time-domain curve and time-frequency curve are presented in Fig. 6(c) and (d), respectively. These curves exhibit rapid amplitude attenuation, unstable fluctuations, and an ambiguous periodic pattern. At the start, there exist numerous high-frequency components. This indicates the superposition of the respiration signals from multiple targets, rendering the respiration signals impure. Later, the signal amplitude is small, and the fluctuations are irregular, making it difficult to identify the respiration rate. At this time, the time-domain signal waveform has severe aliasing, and it is impossible to effectively extract the respiration signals of each target.

Subsequently, we controlled the DPM to direct the Wi-Fi transmitting beam toward the 30° and -20° directions, respectively, and recorded the measured CSI time-domain signals and time-frequency curves. The time-domain and time-frequency curves of the target respiration information in the 30° and -20° directions are shown in Fig. 6(e)–(h), respectively. The respiration rate of target T1 in the 30° direction is detected to be 20 BPM, while that of target T2 in the -20° direction is 15 BPM. Compared with the mixed respiration signal waveform when the DPM is not working, these two signals have higher amplitude stability and slower attenuation, and are smoother and more regular overall. The waveform has a clear periodicity and less noise, and the

respiration rate is easier to identify, avoiding the previous phenomena of severe initial fluctuations and later signal weakening. Moreover, these two waveforms maintain good signal quality at different angles, and the measurement stability and anti-interference ability are significantly improved.

The experimental results show that the DPM in the static operating mode can significantly enhance the respiration signals of the target in the specified direction while effectively suppressing the interference from other directions. Compared with the state when the metasurface is not working, the respiration waveform sensed by CSI has a higher amplitude, a more stable periodicity, and significantly improved signal quality. When the metasurface is not used, the signal decays rapidly and has a large amount of noise, making it difficult to extract effective information in the respiration frequency band. This experiment successfully validates the effectiveness of the metasurface technology in enhancing the target respiration signals and suppressing the interference of respiration signals from irrelevant directions.

D. Multiperson Respiration Detection Under Dynamic Scanning Mode

The above-mentioned experiments verify that, in the static operating mode, the DPM can achieve directional respiration perception and reduce the respiration interference from other directions. Subsequently, we configured the DPM to the dynamic scanning mode. The experimental environment configuration is shown in Fig. 5(a). Four volunteers are positioned at directions of -40° , -10° , 20° , and 40° , each at a distance of 3 m from the center of the DPM. According to the optimized DPM space-coding scheme, the DPM controls the Wi-Fi beam to continuously switch in the azimuth from -50° to 50° (a total of 11 subbeams with an adjacent angle interval of 10°). The CSI packet-sending rate of the Wi-Fi system is 2 ms, and the real-time switching rate of the DPM is 100 Hz. The system can quickly scan different directions and simultaneously capture the respiration information of targets in different directions, improving the real-time performance of perception.

First, we verified the respiration-sensing ability of the original Wi-Fi system in a multiperson scenario when the DPM was in the off state, and Fig. 7(c) and (d) show the CSI time-domain and time-frequency curves. The time-domain signal shows distinct amplitude fluctuations and irregular traits. This reflects the superposition of multiple people's respiration signals, making it arduous to directly extract effective respiratory rhythms. This indicates that it is extremely challenging to completely distinguish the respiration frequencies of various individuals, and a clear-cut separation of targets cannot be achieved. Under the traditional Wi-Fi respiration-sensing method, there are significant signal interference and separation difficulties in CSI respiration sensing in a multitarget scenario.

Subsequently, the DPM operates in the dynamic scanning mode, and the experimental schematic is presented in Fig. 7(b). During the experiment, the DPM divides the CSI of the Wi-Fi transmitter into multiple subsignals in the time domain and modulates the radiation directions of the subsignals for space redistribution. The Wi-Fi signal is reflected

by the DPM and dynamically scanned in space according to the time slices. The signals in a specific direction are enhanced within the corresponding time slices, while the signals in other directions are suppressed. The Wi-Fi Rx receives and extracts the CSI data in different time slices through sampling and then recombines them to restore the human target signals in each angle interval. The respiration time-domain waveforms and time-frequency curves in the -40° , -10° , 20° , and 40° directions perceived by it are shown in Fig. 7(e)–(l), respectively. In the time-domain diagram, the respiration signal waveforms at different angles show high stability and regularity.

The signal amplitude variations extracted in each direction are smooth, featuring clear periodicity without obvious interference. Compared to when the DPM is off, it can effectively separate respiration signals in multiple directions. The signals' observability and quality improve significantly. In the time-frequency-domain diagram, the respiration signals at each angle are concentrated in the corresponding frequency bands, and the frequency lines are clear and nonoverlapping, which indicates that the DPM can effectively suppress other respiration signals from nontarget directions, thereby significantly improving the resolution of the signals in the target direction. Compared with the frequency overlap and aliasing problems when the DPM is not used, the DPM successfully achieves accurate discrimination of respiration signals in different directions by our proposed system.

V. DISCUSSION

In this section, we explore the practical implications of the proposed method's applicability and effectiveness.

Maximum Supportable Targets Before Performance Degradation: The corresponding detection beam along this direction has the gain of

$$G(\theta) = N^2 \cdot \text{sinc}^2\left(\frac{N\pi d}{\lambda} \sin \theta\right) \quad (12)$$

where N is the column number of the DPM, d is the column spacing, and λ is the operating wavelength. In our experiment, $N = 8$, $d = 2.5\text{cm}$, $\lambda \approx 5.3\text{GHz}$. For sufficient signal separation $G(\theta_1)/G(\theta_2) \geq 10\text{dB}$ between the target angle θ_1 and other angle θ_2 , the formula shows the angular offset must exceed 15° . Since the proposed system divides the space into 10° intervals, the two targets need to be spaced at least 20° apart. Therefore, the DPM's configuration allows the system to support up to 7 targets simultaneous detections before performance degradation.

System Performance Under Nonstationary Conditions: Although human movement and posture changes may cause signal fluctuations, the metasurface array captures dynamic features in the time-frequency domain. Transient movements or postural adjustments may induce short-term fluctuations in respiratory signals, but when their spectral characteristics do not overlap with the respiratory frequency band, they do not substantially interfere with respiration detection.

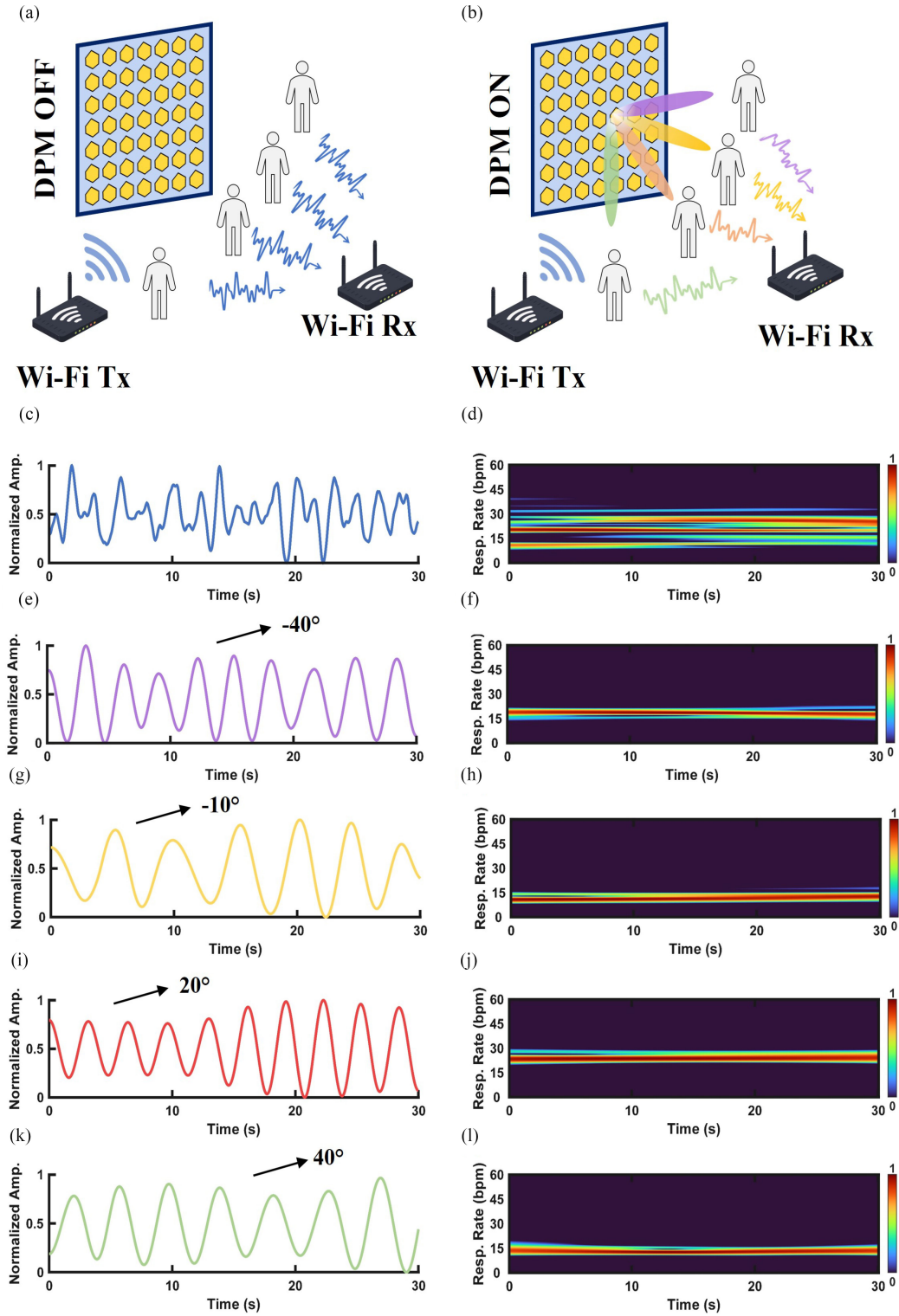


Fig. 7. DPM-aided Wi-Fi respiration detection for four people in Nonworking and dynamic scanning mode. (a) and (b) are schematics of the two states, respectively. (c) and (d) are the respiration signals and the spectrogram in the nonworking state, respectively. (e)–(l) are the respiration signals and spectrograms of the four targets extracted at -40° , -10° , 20° , and 40° .

Scalability Analysis: The angle resolution and beamwidth $\Delta\theta$ can be expressed as

$$\Delta\theta \approx \frac{0.886\lambda}{Nd \cos \theta} \quad (13)$$

where λ is the wavelength, N is the column number, and d is the column spacing. The system's directional support

capacity is constrained by the scale of the metasurface array 8×8 , where physical limitations of beamwidth and angle resolution in the current configuration define the upper bound of multitarget detection. Scalability positively correlates with the number of array elements N : increasing the array scale theoretically narrows the beam width. It enhances angle

resolution proportionally, allowing for the linear expansion of supportable detection directions and target capacity. However, hardware-level tradeoffs exist inherently: higher angular resolution compresses the detection range, while phase distortions induced by target motion in dynamic scenarios significantly degrade detection accuracy.

This study focuses on validating the architecture of DPM-assisted Wi-Fi respiratory detection. Experiments conducted in controlled indoor environments effectively minimize external interference, demonstrating the feasibility of multitarget respiratory monitoring with the proposed hardware setup. Challenges, such as multipath interference, nonline-of-sight (NLOS) obstruction, and co-channel signal interference remain common across wireless sensing technologies, presenting clear and promising directions for future exploration within the framework of the proposed architecture.

Future research will explore adaptive multipath cancellation algorithms in conjunction with the spatial modulation capabilities of metasurfaces to enhance system robustness in complex environments. On the hardware side, increasing the number of meta-atoms and optimizing structural designs are expected to improve angular resolution and spatial perception. On the algorithmic side, exploration will include machine learning-based adaptive beamforming and CSI denoising with dynamic target analysis, enabling more accurate extraction of vital signs and real-time tracking of moving individuals.

VI. CONCLUSION

In this article, we proposed a Wi-Fi respiration detection approach assisted by DPM. We designed and constructed a respiration detection system based on a 2-bit DPM, which utilizes the time and space modulation capabilities of the DPM to assist the Wi-Fi system, enabling the detection of respiratory signals from multiple people. First, we analyze the problem that spectrum aliasing and interference render existing Wi-Fi-based respiratory sensing methods ineffective when there are multiple targets in the same environment. Subsequently, we leverage the time and space modulation capabilities of the DPM to slice the CSI at different time slots and modulate the radiation directions of the CSI slices for space redistribution. Finally, at the receiving end, the CSI is recombined to restore the CSI sensing signals at different angles. Experiments demonstrate that this method can effectively separate and detect the respiratory signals of four people. In the future, our goal is to conduct further research on multiperson vital signs sensing with the help of the DPM, especially in scenarios involving moving individuals, such as walking or running. This will further expand its potential applications in various real-world environments.

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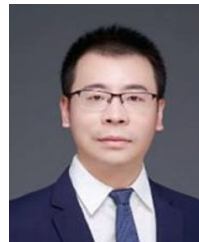
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